Parameter	S over the swept range	No cascade	S range	ΔS
PERC2 settings of facility A (first wave)				
T_{ccease} (s)	0.01 … 0.3	≥ 0.13		81
F_{ccease}	0.1 … 1	≤ 0.75		80
T_{recon} (s)	0 … 5	—		23
T_{delay} (s)	0 … 1	—		12
V_{Lcease} (pu)	0.3 … 0.95	—		0
PERC2 settings of facilities B5, B8 (survivors)				
T_{VHC1} (s)	0.5 … 3	≥ 1.2		70
F_{ccease}	0.05 … 1	—		50
T_{ccease} (s)	0.02 … 0.3	≤ 0.08†		41
T_{delay} (s)	0 … 2	—		22
T_{LreconH} (s)	0.05 … 5	—		0
T_{rampUsed}, HV (s)	0.5 … 20	—		0
Load response of A, B5, and B8				
K_{vq} (T_{vq} = 0.1 s)	0 … 20	≥ 20		71
K_{vq} (T_{vq} = 1 s)‡	0 … 20	≥ 1		57
K_{vp}	0 … 1	—		25
K_{droop}	0 … 40	—		24
n_Q	0 … 3	—		22
n_P	0 … 2	—		21

No cascade: nearest setting to the case file without a load-caused HV cascade (—: none).

Parameter	S over the swept range	No cascade	S range	ΔS
Facility design				
PF-bank ratio k	0 … 2	≤ 1		95
Size of A (MW)	10 … 60	≤ 40		91
Q/P (× case file)	0 … 1.13	≤ 0.75		86
Size of B5 (MW)	5 … 30	—		46
Size of B8 (MW)	5 … 30	—		25
System side (for reference)				
Clearing (cycles)	3 … 9	≤ 4		97
Pre-fault V_{PCC} (pu)	none … 1.05	≤ 1.04		80
Exciter K_{A}	5 … 46	≤ 5		75
Exciter T_{E} (s)	0.2 … 1	≤ 0.2		70
Constant-P share	0 … 0.6	—		15

Cell color: cascade index $S = 50I_{\text{c}} + 25I_{\text{w}} + 25I_{\text{e}}$ (0-100)

fault only, 0 · first wave only, 31.4 · case file, 80.5

0 … 100

Cell marks: case-file setting · load-caused HV cascade

S range: below and above the case-file S

†B5 and B8 cease in the fault. ‡Sustained volt-var variant, not in the case file.